Metrics in "Classification"

Problem → Classification

A,B=0.5

Class Labels

Probability

1. Confusion Matrix
2. FPR (Type I error)
3. FNR (Type II error)
4. Recall (TPR, Sensitivity)
5. Precision (+ve prediction value)
6. Accuracy
7. F Beta
8. Cohen Kappa
9. ROC Curve, AUC Score
10. PR curve.

① Confusion matrix

Predicted Value	1	0	← Actual Value.
1	True +ve	False +ve	→ Type I error ⇓ False +ve Rate FPR
0	False -ve	True -ve	

Type 2 error

False -ve Rate

FNR.

$$FPR \Rightarrow \frac{FP}{FP + TN}$$

$$\Rightarrow Accuracy \Rightarrow \frac{TP + TN}{TP + FP + TN + FN}$$

say in a dataset

$\boxed{A \rightarrow 900 \ , \ B \rightarrow 100}$

↳ using Accuracy matrix on an imbalanced dataset

- Recall → Out of all +ve values (TP, FN) how many
(TPR) did our model predict correctly

$$\frac{TP}{TP+FN}.$$

• Precision → Out of total predicted +ve results, how
(+ve Pred^s Value) many did we ~~ac~~ actually predict right

$$\frac{TP}{TP+FP}.$$

say in confusion matrix,
say Cancer-detection.

		0
0	TP	FP
1	FN	TN

in these cases we use Recall.

↓↓ → Since In 1 column of FN -
if our model predicted
falsely Negative (Person doesn't have Cancer)
It would be a disaster

- If FN imp → Use Recall
- If FP imp → Use Precision

→ F - Beta.

$$\text{F-Beta score} = (1+\beta^2) \cdot \frac{\text{Precision} \times \text{Recall}}{\beta^2 * \text{Precision} + \text{Recall}}$$

$\beta = 1$ F_1 score

$\beta = 0.5$ $F_{0.5}$ Score.

If $\beta = 1$

$$2 \cdot \frac{P \times R}{1 \times P + R} \Rightarrow \left\{ \frac{2PR}{P+R} \right\} \rightarrow \text{Harmonic Mean.}$$

If FP & FN are imp $\beta = 1$

FP is imp. β -- (β val dec)

FN is imp. β ++ (β val inc).

If FN impact is great then ↓

$0 < \beta < 10$